

Verdict-style prompts induce moral over-resolution in large language models

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Abstract

Who was at fault? Who acted unfairly? Who was in the wrong? These are questions people ask large language models. Source-community judgments on such cases can range from near-unanimous to deeply divided. Yet verdict-style answers give one label, and that format can make the chosen answer appear to carry more consensus than the source-community votes support. In a preregistered audit of five LLMs, models evaluated matched ethical scenarios in distribution, verdict/agreement and repeated-sampling modes. In verdict/agreement mode, models reported more source-community agreement for their chosen label than the recorded votes showed — and more than each model’s own distribution-mode probability for that same label, holding model, item and label constant across formats. Repeated forced-choice outputs were also more concentrated than source-community votes, with the clearest effect in divided cases. These results identify a format-specific reliability failure in uncertainty communication: a model that assigns probability across labels in one format can make the same case appear more settled in another.

People now ask large language models to weigh in on everyday conflicts — who was at fault, who acted unfairly, who was in the wrong. But source-community judgments on such cases do not always converge: votes can range from near-unanimous to deeply divided. A verdict-style answer nonetheless gives one label accompanied by an estimate of how much the community would agree. When that estimate exceeds what the votes show, the format communicates more consensus than the evidence supports — and that is the reliability concern this study examines.

Asking for one answer changes what a model communicates about its uncertainty. Given the same case, a model asked to distribute probability across labels may assign substantial weight to several; asked for one label and an agreement estimate, the same model may choose one label and report strong source-community agreement. Those two answers came from the same model on the same case; the prompt mode changed, not the case or the model. When the same case produces different apparent levels of source-community agreement depending on how the question is framed, the verdict-style format can make the case appear more settled than the source-community votes indicate.

Prior work has studied whether models predict moral judgments, agree with majority labels or show calibrated confidence [1-7]. These studies ask how well model outputs match a reference distribution from one format. This study asks a distinct reliability question: when the same model answers the same case in two elicitation formats, do the resulting outputs communicate consistent levels of source-community agreement? Unlike standard calibration, this comparison is internal to the model’s own outputs; it does not require the distributional estimate to have recovered the source-community distribution perfectly.

The failure mode this study tests is format-induced moral over-resolution: a verdict-style prompt can make a divided case appear more settled than the source-community votes (or the model’s own distributional answer) would indicate. Three matched-item endpoints test whether and how much verdict-style outputs over-resolve source-community disagreement. Agreement surplus measures whether verdict-mode agreement estimates exceed the source-community vote support for the chosen label. The distribution-agreement gap is the central cross-format test: it measures whether those estimates also exceed the same model’s own distribution-mode probability for the same label, comparing the same model, item and chosen label across elicitation formats. A positive gap is a cross-format inconsistency within the model’s own outputs: the same model reports more apparent source-community agreement in verdict mode than its distributional answer assigns to that same label. Sampling compression measures whether repeated forced-choice outputs are more concentrated than the source-community vote distribution.

The study uses SCRUPLES Anecdotes, a corpus of everyday ethical situations with recorded votes on who was in the wrong [4]. "Source community" here refers to the annotator population represented in SCRUPLES; the resulting vote distributions are reference data from that data-generating process, not moral truth, universal norms or representative population estimates. In this preregistered audit, five LLMs evaluated matched SCRUPLES items in distribution mode, descriptive verdict/agreement mode and repeated-sampling mode (Fig. 1). Low-consensus items — cases with a majority label but substantial remaining disagreement in the source-community votes — were the primary confirmatory subset. Diffuse/no-clear-consensus items were analysed separately as secondary evidence; high-consensus items served as a reference condition. SCRUPLES provides measured disagreement in everyday moral judgments; the study’s contribution is an evaluation of model-output reliability under format change.

1 Results

1.1 Endpoint logic and target set

We tested three ways verdict-style answers can over-resolve source-community disagreement — agreement surplus, distribution-agreement gap and sampling compression — using matched item-model comparisons. Low-consensus items (cases with a majority label but substantial remaining source-community disagreement) were the primary confirmatory subset; diffuse/no-clear-consensus items were analysed separately as secondary evidence; high-consensus items served as a reference condition. The model roster and study scale are reported in Table 1 and Methods; 47,500 target calls produced 47,432 primary-valid outputs.

1.2 Verdict prompts inflate apparent agreement

Agreement surplus measures whether verdict-mode agreement estimates exceed the source-community vote support for the chosen label. In the low-consensus primary subset, agreement surplus was 0.370931 (95% CI, 0.361206-0.381114; Holm-adjusted $P = 0.0015$ (bootstrap floor); $n = 3,749$ item-model rows; Fig. 2 and Table 2): verdict-mode estimates exceeded source-community vote support for the chosen label by approximately 37 percentage points. The accompanying agreement estimate can make the chosen label appear more widely supported than the source-community votes show.

The same direction appeared outside the primary subset. Diffuse/no-clear-consensus items showed a larger secondary estimate: agreement surplus was 0.435619 (95% CI, 0.427337-0.444341; $n = 2,500$). High-consensus items also showed positive agreement surplus, 0.169409 (95% CI, 0.145651-0.194624; $n = 1,750$), with a low-consensus-minus-high-consensus contrast of 0.201523 (95% CI,

0.174835-0.228568). High-consensus items therefore served as a positive reference condition, while the larger low-consensus surplus shows greater reliability concern where source-community agreement was weaker.

1.3 Verdict agreement exceeds the model’s distributional estimate

The distribution-agreement gap is the study’s central within-model coherence test: it compares each model’s verdict-mode agreement estimate with the same model’s distribution-mode probability for the same item and chosen label. A positive gap means the verdict/agreement format reported more estimated source-community agreement for the chosen label than the same model assigned to that label in distribution mode.

This comparison is internal to the model’s own outputs: it holds model, item and chosen label constant while the elicitation format differs, and therefore does not require the distribution-mode estimate to be perfectly calibrated to the source-community distribution.

In the low-consensus primary subset, the distribution-agreement gap was 0.232687 (95% CI, 0.224387-0.241341; Holm-adjusted $P = 0.0015$ (bootstrap floor); $n = 3,736$; Fig. 3 and Table 2): for the same item and chosen label, verdict/agreement mode reported more agreement than the same model’s distribution-mode probability supported.

Diffuse items showed a secondary gap of 0.226556 (95% CI, 0.216740-0.237164; $n = 2,494$). High-consensus items also showed a positive gap of 0.182842 (95% CI, 0.169762-0.196217; $n = 1,745$), with a low-consensus-minus-high-consensus contrast of 0.049845 (95% CI, 0.033774-0.065329). High-consensus items remained a positive reference condition; the larger low-consensus contrast indicates a greater coherence concern when source-community votes were more divided. Within the three-endpoint design, this gap provides the cleanest within-model coherence comparison: the mismatch is not between model outputs and source-community votes but between two outputs from the same model for the same case and label. Its positive value across all disagreement bins shows that cross-format coherence failure extends beyond divided cases; agreement surplus and sampling compression show that the magnitude of over-resolution is greatest where source-community votes are most divided.

1.4 Repeated forced-choice outputs compress source-community disagreement

The first two endpoints concern stated agreement estimates. The third asks whether the same concentration pattern appears when models are asked only to choose labels repeatedly, without stating agreement. Sampling compression is source-community entropy minus the entropy of repeated forced-choice outputs on the five-label scale; positive values mean outputs were more concentrated across labels than source-community votes for the same items (see Methods for the collection protocol).

In the low-consensus primary subset, sampling compression was 1.264638 bits (95% CI, 1.218425-1.309054; Holm-adjusted $P = 0.0015$ (bootstrap floor); $n = 750$ item-model summaries; Fig. 4 and Table 2). Diffuse items showed stronger secondary compression, 1.507379 bits (95% CI, 1.452200-1.562768; $n = 500$). High-consensus items also showed positive compression, 0.563708 bits (95% CI, 0.495005-0.633547; $n = 350$). The low-consensus-minus-high-consensus contrast was 0.700930 bits (95% CI, 0.618287-0.783543).

Under the study’s repeated forced-choice protocol, outputs across fresh calls were more concentrated than source-community votes, with larger compression in the low-consensus primary subset than in the high-consensus reference condition. This endpoint extends the finding beyond stated agreement: concentration appears in label choices as well. Repeated forced-choice outputs under

this protocol measure label diversity across fresh calls rather than stochastic sampling from an internal model distribution.

1.5 Pattern across models and output validity

All five evaluated models showed positive low-consensus agreement surplus, positive low-consensus distribution-agreement gap and positive low-consensus sampling compression. Model-level low-consensus agreement surplus ranged from 0.326454 to 0.445325. Low-consensus distribution-agreement gaps ranged from 0.215187 to 0.271943. Low-consensus sampling compression ranged from 0.920001 to 1.408416. Leave-one-model-out summaries preserved the direction of the aggregate pattern for all three endpoints. These summaries support a consistent directional pattern across the five evaluated models; five models from the same deployment era cannot establish generalization across model families, training paradigms or deployment contexts. Provider, route and model-lineage comparisons remain outside the analysis.

Output validity was high: of 47,500 outputs, 47,432 were primary-valid (validity rate = 0.998568), comprising 38,790 strict-schema outputs and 8,642 extracted-JSON outputs. The minimum model/model primary-valid rate was 0.969, with zero refusals, zero off-schema labels and no repaired outputs. Detailed validity counts by model, prompt mode and invalid status are in Extended Data Table 1 and Extended Data Fig. 3.

1.6 Robustness and diagnostics

The directional pattern of the three low-consensus endpoint results held across robustness checks that recomputed the endpoints under alternative validity, reference-distribution and item-composition choices.

A strict-valid-only analysis restricted the data to outputs satisfying the strict schema. The direction of all three low-consensus endpoints held: agreement surplus 0.376030, distribution-agreement gap 0.223867 and sampling compression 1.268188.

The source-community reference distribution was checked against alternative constructions from SCRUPLES vote counts. Low-consensus endpoint effects were recomputed using raw proportions, Jeffreys smoothing and Laplace smoothing, and positive effects were preserved across all three. These checks addressed reference-distribution construction, not the calibration of distribution-mode outputs to those distributions.

Item-composition sensitivity preserved the directional endpoint pattern in high-annotation-only analyses and after excluding **info**-majority or high-**info** items.

Distribution-mode diagnostics confirmed that model outputs varied by item and disagreement bin. Source entropy was higher in diffuse items (mean 1.955481 bits) than in high-consensus items (0.823596 bits). Model distribution entropy also varied by bin, with higher entropy in diffuse (1.515785 bits) and low-consensus items (1.476755 bits) than in high-consensus items (1.200675 bits). These analyses, summarized in Extended Data Fig. 1, are diagnostic rather than distribution-prediction or perfect-calibration tests.

The paraphrase audit provided aggregate surface-form evidence in paraphrased items (Extended Data Fig. 2), with 2,500 target calls and 2,495 valid outputs. In paraphrased low-consensus items, agreement surplus was 0.366821 (95% CI, 0.334632-0.397075) and the distribution-agreement gap was 0.212124 (95% CI, 0.192023-0.231651). In paraphrased diffuse items, the corresponding estimates were 0.418159 (95% CI, 0.396906-0.443057) and 0.200502 (95% CI, 0.174963-0.225410).

Within the limited matched original-vs-paraphrase overlap, chosen-label stability was 0.655 in low-consensus rows and 0.467 in diffuse rows. With 29 matched low-consensus rows and 15 matched

diffuse rows, this component provides aggregate surface-form evidence rather than a definitive paired original-vs-paraphrase test.

2 Discussion

Under the tested prompt formats and evaluated models, verdict-style answers made divided cases appear more settled than either the source-community votes or the same model’s own distributional answer would indicate. The distribution-agreement gap provides the cleanest within-model coherence test: for the same model, item and chosen label, verdict-mode agreement estimates exceeded the model’s own distribution-mode probability for that label — a within-model comparison that holds model, item and label constant. The discrepancy is not a calibration error against source-community votes; it is an inconsistency between two outputs from the same model on the same case. This gap, together with positive agreement surplus and sampling compression in the same low-consensus setting, shows that the apparent level of source-community agreement a model communicates can depend on the elicitation format.

The finding has implications for how LLM reliability is evaluated. Majority-label accuracy can reward a model for selecting the most common label without indicating how that model estimates source-community agreement in verdict mode. Distribution-mode uncertainty tests are similarly incomplete on their own: a model can produce non-degenerate distributional outputs while also reporting higher apparent source-community agreement in verdict mode for the same item and label. Prior work shows that benchmark conclusions can depend on prompt formatting [8]; the present results extend that concern to whether agreement estimates remain consistent when the interface changes from distributional to verdict-style.

The findings also bear on interface design. When a model is prompted for a verdict on a divided case, the accompanying agreement estimate can substantially exceed what the source-community votes and the model’s own distributional answer would support for that label. Within the domain tested here, verdict-mode agreement estimates were not interchangeable with distributional probability estimates for the same case and label. Whether interface changes would reduce that gap is a question outside this study.

The findings are bounded by one source-community dataset, five evaluated models, a finite set of prompt templates and a source community that is not a universal or representative population sample.

The paraphrase audit provides aggregate surface-form evidence; it did not test training-set membership; contamination was not directly measured; and the small matched original-vs-paraphrase overlap limits paired interpretation. Item familiarity could affect absolute label probabilities, though the distribution-agreement gap compares the same model’s outputs across formats for the same item rather than against an external calibration target. The study leaves user effects — including effects on perceived social consensus, confidence and decision-making — as an open empirical question.

The implication for model-output reliability is direct. Testing models only in distribution mode can miss over-resolution in verdict mode: a model can produce non-degenerate distributional outputs while reporting higher apparent source-community agreement in verdict mode for the same case and label. Reliability evaluations that compare verdict-mode agreement estimates against the same model’s distributional outputs for matched items can identify that discrepancy. An evaluation limited to one format can make distributional uncertainty appear adequate in that format while missing how the same model behaves when asked for a verdict.

3 Methods

3.1 Preregistration and analysis scope

The study was preregistered on OSF as a target-scoped computational audit. It used existing SCRUPLES source-community judgments and newly collected LLM API outputs; no new human participants were recruited. The analysis uses the frozen 50k target-scoped exports from `post_run/analysis_exports/50k/`.

Operational milestones tracked monitoring and release discipline. The three primary endpoint definitions and the low-consensus confirmatory subset were specified in the preregistered protocol before any confirmatory data collection; they remained unchanged through the final 50k analysis. Subsequent presentation choices did not alter the frozen numerical outputs.

The primary confirmatory endpoints were agreement surplus, distribution-agreement gap and sampling compression in low-consensus items. Diffuse/no-clear-consensus items were theoretically important secondary evidence and remained separate from the primary endpoint definition. Combined low-consensus and diffuse aggregates were used only as descriptive divided-item summaries where needed.

3.2 Dataset, eligibility and label schema

The source dataset was SCRUPLES Anecdotes, a corpus of everyday ethical situations with source-community judgments about who was in the wrong [4]. Items were eligible when they had usable situation text, available source-community counts, canonical label mapping and no data-integrity issue preventing item-level analysis. These judgments were treated as reference distributions from the SCRUPLES data-generating process, not as moral truth, universal norms, representative population estimates or deployment-ready normative labels.

The canonical label schema was:

```
author      = the author/narrator was in the wrong
other       = the other party was in the wrong
everybody   = everybody/both sides were in the wrong
nobody      = nobody was in the wrong
info        = not enough information/cannot tell
```

The `info` label was retained in the primary distributions because removing it would alter the observed disagreement structure.

3.3 Source-community distributions, bins and item allocation

For each item, source-community vote counts were converted to primary source-community distributions using Jeffreys-smoothed Dirichlet posterior means with $\alpha = 0.5$:

```
theta_i | c_i ~ Dirichlet(c_i + 0.5)
primary_source_distribution_i = E[theta_i | c_i]
```

Robustness checks used raw proportions and Laplace smoothing with $\alpha = 1.0$.

Items were stratified by top-label support:

```
high consensus:      top-label support >= 0.80
moderate consensus:  top-label support >= 0.65 and < 0.80
```

low consensus: top-label support ≥ 0.50 and < 0.65
diffuse/no-clear-consensus: top-label support < 0.50

Item selection and label-order assignment used the recorded milestone seed 20260615. Within each disagreement bin, eligible items were selected according to the frozen allocation plan and component-specific target counts; the released target manifests record the resulting item IDs. Beyond the eligibility rules above, target-list membership was determined by the frozen allocation plan and recorded seed rather than by inspection of model outputs.

Low-consensus items were the primary confirmatory subset, diffuse/no-clear-consensus items remained secondary, and high-consensus items served as the reference condition.

The core cross-format component used 350 high-consensus, 400 moderate-consensus, 750 low-consensus and 500 diffuse/no-clear-consensus items. The repeated-sampling, paraphrase-audit and normative-certainty components used component-specific target sets. The paraphrase audit used a stratified subset for aggregate surface-form checks.

3.4 Model roster and collection

The amended frozen roster contained five models: `claude-sonnet-4-6`, `deepseek/deepseek-v3.2`, `gpt-5.5`, `grok-4.3` and `qwen/qwen3.7-max`. Each model had 9,500 target calls. API routes and call windows are reported in Table 1, panel a.

Temperature was 0.0 for all exported rows. Top-p was blank/null in the final model roster. Provider and route fields record access provenance and are not analysed as cross-provider group effects. Exact model snapshot dates were not available in the local final roster and are not inferred beyond the recorded first and last call timestamps.

Full API-parameter fields, structured-output settings where applicable, maximum-token settings, retry-policy fields, reasoning-effort fields when present and collection timestamps are reported in the Supplementary Information and exported roster. The Supplementary Information and run manifests document the roster amendment and final completion/analysis provenance. The final analysis uses only the amended frozen five-model roster.

3.5 Prompt modes and request construction

Distribution mode asked each model to estimate the source-community probability distribution over the five labels. Descriptive verdict/agreement mode asked the model to choose one label and estimate source-community agreement for that choice. Repeated-sampling mode elicited ten single-label forced-choice outputs for each repeated-sampling item-model pair. Normative-certainty prompts elicited a separate moral-certainty construct for secondary descriptive analysis.

Requests were fresh and stateless across prompt modes. The verdict/agreement prompt did not include the model’s own distribution-mode answer. Prompt templates, schemas, label order, rendered prompt hashes, model IDs, API route, timestamps and decoding parameters were recorded in run records. Prompts requested structured final outputs and did not request chain-of-thought.

For repeated sampling, the ten outputs for each item-model pair were ten separate fresh calls under the sampling prompt at temperature 0.0. Label order was assigned separately for each prompt assignment using the recorded seed procedure; `sample_id` was part of the local label-order seed. Invalid outputs were handled under the same parsing and validity rules as other components before entropy summaries were computed. The sampling-compression endpoint measures diversity across repeated forced-choice outputs under the recorded prompt and label-order randomization protocol rather than stochastic sampling from an internal model distribution.

For paraphrase analyses, paraphrased versions of a stratified item subset were generated with the study’s paraphrase-generation prompt mode. The paraphrased items were then evaluated with the paraphrased distribution and paraphrased descriptive verdict/agreement modes, using the same response schemas and parsing/validity rules as corresponding original-item prompts. Because matched original-vs-paraphrase coverage was limited, the paraphrase component provides aggregate surface-form evidence rather than a definitive paired original-vs-paraphrase test. This component did not test training-set membership, and contamination was not directly measured.

Rendered prompt hashes were computed after inserting item text and randomized label order. Public materials describe prompt structure and release-safe templates while excluding rendered prompts containing source anecdotes.

3.6 Parsing, validity and exclusions

Strict JSON schema enforcement was used where supported. Otherwise, JSON was extracted from model responses and validated against the required fields and ranges. Outputs were excluded from primary analyses if the response was empty, invalid JSON, out of range, outside the validation tolerance for five-label probability sums, missing required fields, a refusal, an off-schema label or a terminal/API failure without a valid response.

For distribution-mode outputs, all five label probabilities had to be castable to floats and within the closed interval $[0, 1]$. Values outside that interval were marked as **probability_out_of_bounds**.

Probability vectors were accepted when their total fell within the inclusive interval $[0.99, 1.01]$. Accepted vectors whose totals differed from 1.0 by more than $1e-12$ were renormalized to sum to one before downstream use. The $1e-12$ threshold was only the renormalization threshold, not the exclusion tolerance. Vectors with totals below 0.99 or above 1.01 were marked as **probability_sum_error** and excluded from primary analyses. These statuses were assigned by the collection-time parser; the final 50k exporter inherited stored parser validity statuses rather than revalidating probability sums or applying a second tolerance.

The frozen analysis included 47,500 planned, attempted and completed target calls. Run prove-nance recorded 0 API errors, 0 terminal failures, retry rate 0.0, executor status passed and database integrity ok.

Primary-valid outputs were 47,432 of 47,500: 38,790 strict-schema outputs and 8,642 extracted-JSON outputs. Invalid status counts were 52 **probability_out_of_bounds** outputs, 10 **probability_sum_error** outputs, 4 invalid-JSON outputs and 2 empty responses. Refusals and off-schema labels were zero. Repair fields were tracked, but **repair_attempted=0** and **repair_successful=0**; the final 50k analysis contained no repaired outputs. Validity by model and prompt mode is reported in Extended Data Table 1.

Strict-schema support varied across access routes. The robustness analyses included a strict-valid-only subset, which preserved the direction of all three low-consensus endpoints and is reported in Results and the Supplementary Information.

3.7 Endpoint definitions

For each verdict/agreement output, source-community support for the model’s chosen label was computed from the smoothed source-community distribution. Agreement surplus was defined as:

$$\begin{aligned} \text{agreement_surplus} = & \\ & \text{verdict_mode_estimated_source_community_agreement} \\ & - \text{source_support_for_model_chosen_label} \end{aligned}$$

For each item-model pair with both distribution-mode and verdict/agreement-mode outputs, distribution-agreement gap was defined as:

```
distribution_agreement_gap =
    verdict_mode_estimated_agreement
    - distribution_mode_probability_for_same_verdict_label
```

This endpoint compares the same model, item and verdict-selected label across two elicitation formats. It tests output coherence; perfect calibration of the distribution-mode estimate to the source-community distribution is not required.

For repeated sampling, source-community entropy was compared with the entropy of repeated forced-choice model outputs:

```
sampling_compression =
    source_community_entropy
    - repeated_sample_model_entropy
```

Entropy values are reported on the five-label scale in bits. Positive values indicate that repeated forced-choice outputs under the study protocol are more concentrated than the source-community distribution.

3.8 Statistical inference

Primary inference used item-cluster bootstrap confidence intervals with 2,000 bootstrap iterations and seed 20260621. For bin-specific estimates, each bootstrap replicate resampled item IDs with replacement within the bin and retained all relevant model/prompt rows associated with the resampled items.

For contrasts, each bootstrap replicate recomputed the relevant bin means and subtracted the high-consensus reference estimate. This item-level clustering avoids treating item-model rows as fully independent observations.

The three low-consensus primary endpoints were tested using one-sided positive-effect bootstrap P values, Holm-adjusted across the three endpoints. Because 2,000 bootstrap iterations impose a finite floor, adjusted P values are reported as Holm-adjusted $P = 0.0015$.

Contrasts comparing low-consensus or diffuse items with high-consensus items used the same bootstrap framework. Mixed-effects model-ready rows are provided as transparency files; mixed-effects model results are outside the reported analyses.

3.9 Robustness and secondary analyses

Robustness analyses covered five areas: parsing and schema validity, source-community distribution construction, annotation and **info** sensitivity, distribution-mode non-degeneracy, and prompt surface form. Checks included strict-valid-only subsets; raw, Jeffreys-smoothed and Laplace-smoothed source-community distributions; high-annotation-only subsets; exclusion of **info**-majority and high-**info** items; distribution-quality diagnostics with non-API baselines; and aggregate paraphrase analyses.

Distribution-quality diagnostics included Jensen-Shannon divergence, total variation distance, entropy and Brier summaries, with uniform, global-base-rate and source-majority-oracle baselines. These diagnostics assessed non-degeneracy and item sensitivity rather than perfect distribution calibration.

Normative certainty was analysed as a secondary descriptive construct distinct from estimated source-community agreement. No formal recognition audit was run for the final 50k analysis, and contamination was not directly measured.

4 Data availability

Derived analysis files and reproducibility records are available through the OSF project (<https://osf.io/rwhax/overview>), OSF registration home (<https://osf.io/rwhax/registrations>), GitHub repository (<https://github.com/pbriggs/format-induced-moral-overresolution>), the post-50k execution/completion archive (<https://doi.org/10.5281/zenodo.20786461>) and the paper-analysis archive (<https://doi.org/10.5281/zenodo.20789625>). The corresponding release tags are `post-50k-completion-v1` and `paper-analysis-50k-v1`.

Release-safe materials include frozen target lists and run manifests with sensitive details removed where needed; prompt templates and schemas; derived endpoint, robustness, paraphrase, baseline, secondary normative-certainty and distribution-quality summaries; bootstrap intervals, adjusted tests and validity/exclusion summaries; manuscript-table CSVs, figure-ready CSVs and rendered figures. The Supplementary Information and repository documentation list the main source-data and analysis files, and analysis code is described below. The repository license is CC0 1.0 Universal unless superseded by OSF project settings.

Raw SCRUPLES anecdotes, rendered prompts containing anecdote text, newly collected raw model/provider responses, full call ledgers and full run stores containing restricted or mixed material are not redistributed in unrestricted form where doing so could conflict with SCRUPLES/AllenAI terms, model-provider terms, privacy/sensitivity considerations or institutional review requirements. The source SCRUPLES data should be accessed through the original dataset source subject to its terms [4]. Restricted verification materials can be provided to editors or reviewers upon reasonable request under appropriate non-redistribution conditions and applicable third-party terms.

5 Code availability

Analysis and figure-generation code are available at <https://github.com/pbriggs/format-induced-moral-overresolution> and archived under release tag `paper-analysis-50k-v1` with DOI <https://doi.org/10.5281/zenodo.20789625>. The repository includes preprocessing, prompt/schema, parsing/validation, bootstrap-analysis, manuscript-table and figure-rendering code. API credentials and provider-specific secrets are not included.

Local setup and regeneration instructions are provided in `README.md`, `.env.example` and `docs/DATA_SETUP.md`. Offline regeneration uses completed-run data in an authorized local or reviewer environment and does not make new provider calls. Optional provider-call reruns require the rerunning researcher’s own provider credentials and may produce new model outputs rather than exactly reproducing the frozen 50k collection.

6 AI assistance disclosure

The author used AI tools for editorial drafting support, code and checklist support, manuscript organization and revision planning. The author reviewed, verified and takes responsibility for all analyses, claims, interpretations and final manuscript text. No AI tool is listed as an author.

7 Ethics and human-subjects statement

No new human participants were recruited for this computational audit, and the author did not interact with human participants. The study analysed an existing SCRUPLES dataset of source-community judgments and newly collected model outputs. No formal institutional review board or ethics-committee determination was obtained for this independent computational study. Raw anecdotes, rendered prompts and raw model/provider responses are treated cautiously because they may contain sensitive interpersonal content and may be subject to dataset terms, provider terms and privacy considerations.

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9 Competing interests

The author declares no competing interests.

10 Author contributions

P.B. conceived the study, designed the protocol, implemented the data-collection and analysis pipeline, conducted the analyses, interpreted the results, prepared the figures and tables, and wrote and revised the manuscript.

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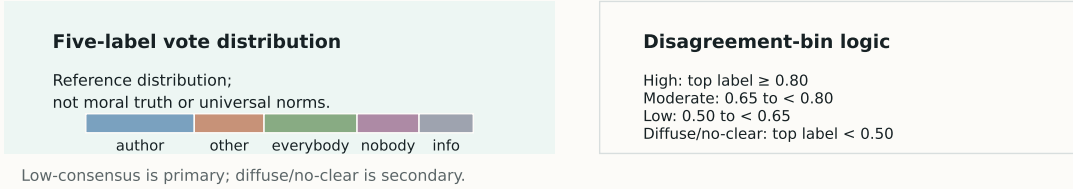
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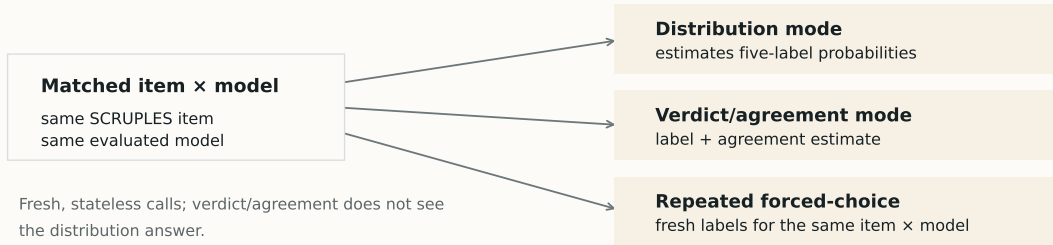
7. Zaim bin Ahmad, M. S. & Takemoto, K. Large-scale moral machine experiment on large language models. *PLoS One* **20**, e0322776 (2025). <https://doi.org/10.1371/journal.pone.0322776>
8. Sclar, M., Choi, Y., Tsvetkov, Y. & Suhr, A. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. *International Conference on Learning Representations* (2024). <https://openreview.net/forum?id=RIu5lyNXjT>

Study design for testing cross-format uncertainty transfer

a Source-community distributions and bins



b Same item $\times$ same model across prompt formats



c Primary endpoint definitions

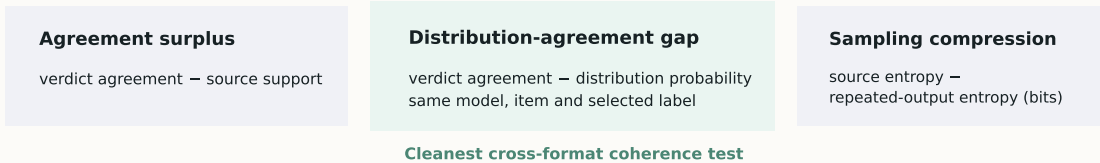


Figure 1: Fig. 1 | Study design for testing cross-format agreement consistency. SCRUPLES source-community vote distributions were grouped by source-community disagreement bin and evaluated in matched item-model comparisons across prompt formats. Distribution mode elicited five-label source-community probability estimates; descriptive verdict/agreement mode elicited one label and an estimated source-community agreement value; repeated-sampling mode elicited repeated forced-choice outputs for the same item-model pair. The three primary endpoints are agreement surplus, distribution-agreement gap and sampling compression. Together, they compare source-community support, distribution-mode probabilities, verdict/agreement estimates and repeated-output entropy. Paraphrase-audit results provide supporting aggregate surface-form evidence, and normative-certainty results are secondary descriptive evidence; neither component is a primary endpoint. This schematic is a design overview; quantitative endpoint results are reported in Figs. 2–4 and Table 2.

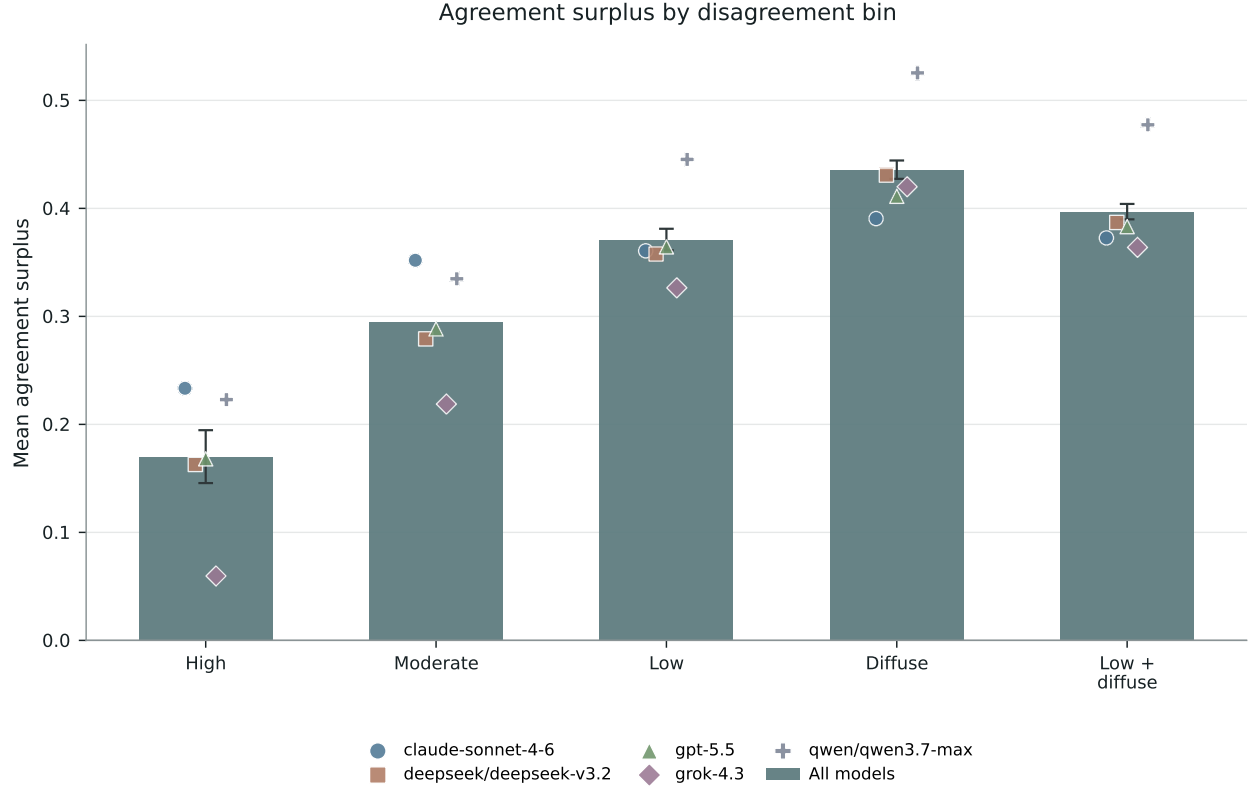


Figure 2: **Fig. 2 | Verdict prompts inflate apparent source-community agreement.** Agreement surplus by source-community disagreement bin and model. Bars show aggregate means across the evaluated models; overlaid points show model-level means. Positive values mean that the verdict-mode estimated source-community agreement exceeds source-community support for the model’s chosen label. In the low-consensus primary subset, mean agreement surplus was 0.370931 (95% item-cluster bootstrap CI, 0.361206-0.381114; $n = 3,749$ item-model rows; Holm-adjusted $P = 0.0015$ (bootstrap floor)).

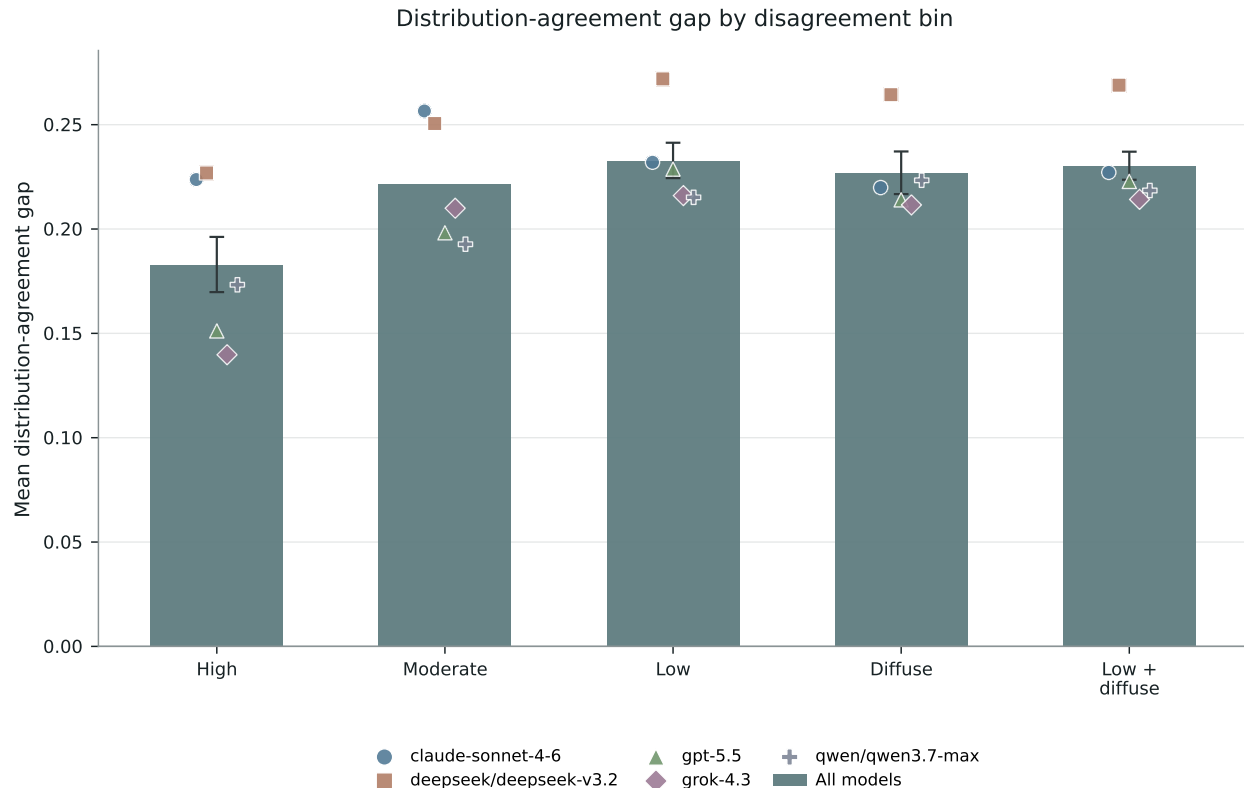


Figure 3: **Fig. 3 | Verdict prompts overstate agreement relative to the same model’s distribution-mode estimate.** Distribution-agreement gap by source-community disagreement bin and model. Bars show aggregate means across the evaluated models; overlaid points show model-level means. Positive values mean that the verdict-mode agreement estimate exceeds the same model’s distribution-mode probability for the same item and verdict-selected label. This endpoint is the cleanest within-model cross-format coherence test because the model, item and selected label are fixed while the elicitation format changes; it does not require perfect calibration of distribution-mode estimates to the SCRUPLES source-community distribution. In the low-consensus primary subset, mean distribution-agreement gap was 0.232687 (95% item-cluster bootstrap CI, 0.224387-0.241341; $n = 3,736$ item-model rows; Holm-adjusted $P = 0.0015$ (bootstrap floor)).

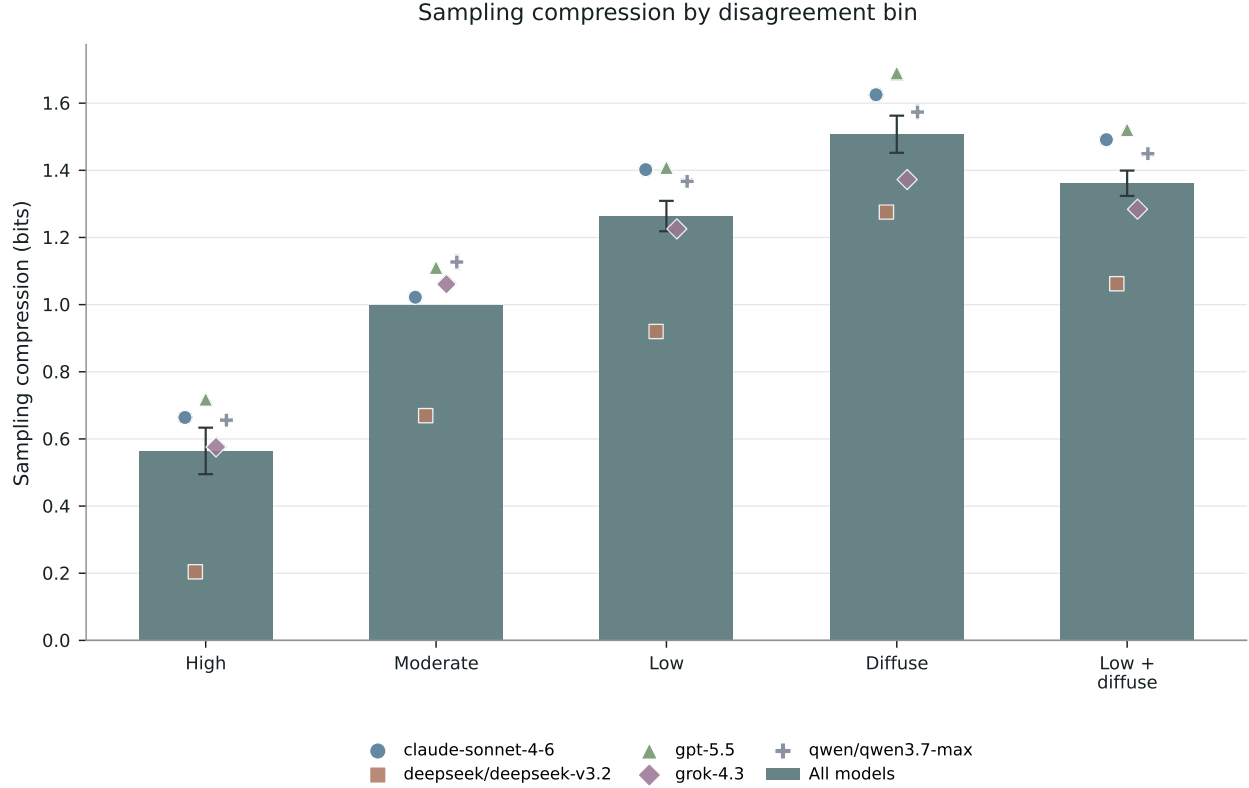


Figure 4: **Fig. 4 | Repeated forced-choice outputs compress source-community disagreement.** Sampling compression by source-community disagreement bin and model. Bars show aggregate means across the evaluated models; overlaid points show model-level means for source-community entropy minus entropy of repeated forced-choice outputs on the five-label scale in bits. Positive values mean that repeated forced-choice outputs are more concentrated than source-community judgments under the study protocol. In the low-consensus primary subset, mean sampling compression was 1.264638 bits (95% item-cluster bootstrap CI, 1.218425-1.309054; $n = 750$ item-model summaries; Holm-adjusted $P = 0.0015$ (bootstrap floor)).

Table 1 | Model roster and target allocation. The roster reports the evaluated model IDs, access routes, collection windows and target calls. Provider/route fields record provenance only; provider-family, route and model-family comparisons are outside the analysis. The allocation table reports the frozen study components used for the 50k analysis.

a, Model roster and collection windows

Model ID	Provider/route	API route	First call UTC	Last call UTC	Target calls
claude-sonnet-4-6	Anthropic	https://api.anthropic.com/v1/messages	2026-06-16T22:25:08.996871+00:00	2026-06-21T15:33:15.162406+00:00	9,500
deepseek/deepseek-v3.2	OpenRouter-compatible route	https://openrouter.ai/api/v1/chat/completions	2026-06-16T22:25:04.892615+00:00	2026-06-21T15:33:02.923140+00:00	9,500
gpt-5.5	OpenAI	https://api.openai.com/v1/responses	2026-06-16T22:25:12.406257+00:00	2026-06-21T15:33:16.432159+00:00	9,500
grok-4.3	xAI	https://api.x.ai/v1/responses	2026-06-17T17:55:15.846572+00:00	2026-06-21T15:32:59.334062+00:00	9,500
qwen/qwen3.7-max	OpenRouter-compatible route	https://openrouter.ai/api/v1/chat/completions	2026-06-16T22:25:15.735026+00:00	2026-06-21T15:33:13.991444+00:00	9,500

Temperature was 0.0 for all exported model-roster rows; top-p was blank/null. Provider/route fields record access provenance and are not analysed as cross-provider group effects; model snapshot dates are not inferred beyond the recorded first and last call timestamps. Total target calls were 47,500.

b, Target allocation by study component

Component	Unique items	Target calls	Manuscript role
Core cross-format	2,000	20,000	H1a and H1b
Repeated sampling	400	20,000	H1c
Paraphrase audit	250	2,500	Aggregate surface-form evidence
Normative certainty	1,000	5,000	Secondary descriptive analysis
Total		47,500	Frozen 50k target-scoped analysis

Table 2 | Primary endpoint estimates, contrasts, confidence intervals and adjusted tests.

Endpoint	Low-consensus primary mean (95% CI)	Holm- adjusted P	Diffuse secondary mean (95% CI)	High-consensus reference mean (95% CI)	Low-high contrast (95% CI)
Agreement surplus	0.370931 (0.361206-0.381114)	0.0015	0.435619 (0.427337-0.444341)	0.169409 (0.145651-0.194624)	0.201523 (0.174835-0.228568)
Distribution-agreement gap	0.232687 (0.224387-0.241341)	0.0015	0.226556 (0.216740-0.237164)	0.182842 (0.169762-0.196217)	0.049845 (0.033774-0.065329)
Sampling compression, bits	1.264638 (1.218425-1.309054)	0.0015	1.507379 (1.452200-1.562768)	0.563708 (0.495005-0.633547)	0.700930 (0.618287-0.783543)

Estimates use the frozen 50k target-scoped analysis. Low-consensus values are the primary confirmatory estimates; diffuse/no-clear-consensus values are secondary evidence and remain separate from the primary endpoint definition; high-consensus values are positive estimates for the reference condition. Confidence intervals are item-cluster bootstrap intervals with 2,000 bootstrap iterations. P values are one-sided positive-effect bootstrap P values, Holm-adjusted across the three low-consensus primary endpoints; with 2,000 bootstrap iterations, Holm-adjusted P values are reported at the finite bootstrap floor of 0.0015.

13 Extended Data and Supplementary Information

The Supplementary Information provides reviewer-facing methodological detail, source-data references and release-safe reproducibility information for the Extended Data figures, Extended Data table and Supplementary Tables. The corresponding source-data files are listed in the Supplementary Information rather than repeated in full in the main manuscript.

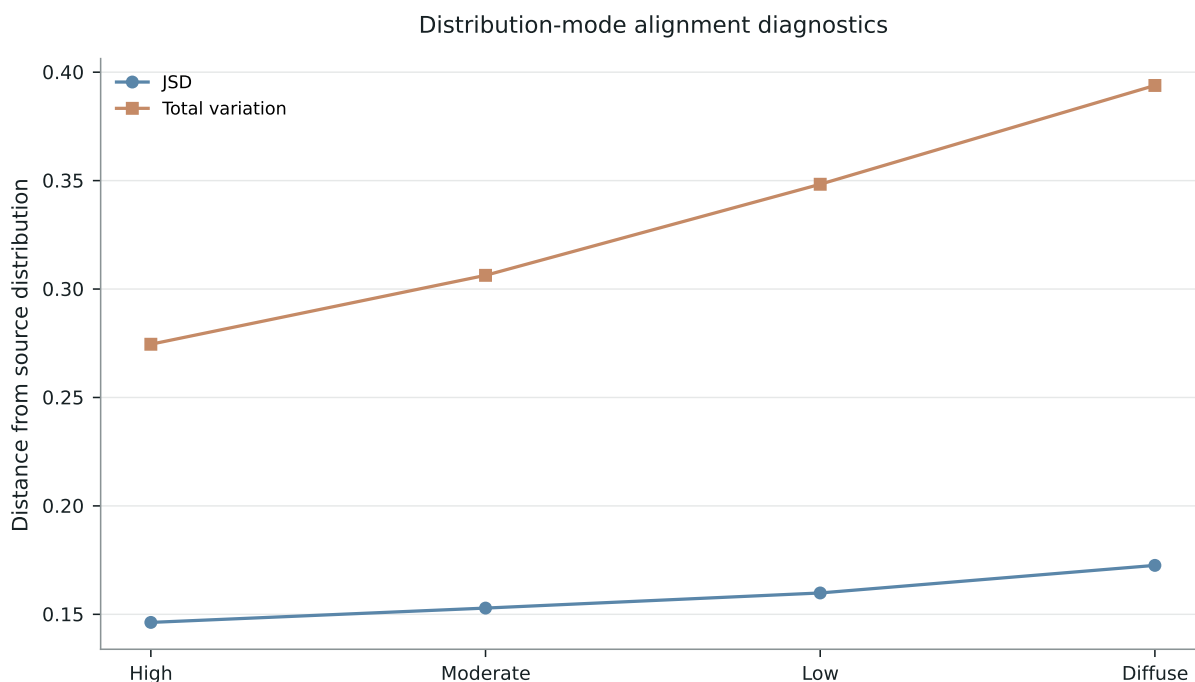


Figure 5: **Extended Data Fig. 1 | Distribution-quality diagnostics.** Diagnostic distances and entropy summaries by source-community disagreement bin and model show that distribution-mode outputs varied by item and bin and were not merely uniform or fixed base-rate responses. This is a diagnostic support analysis, not a distribution-prediction benchmark.

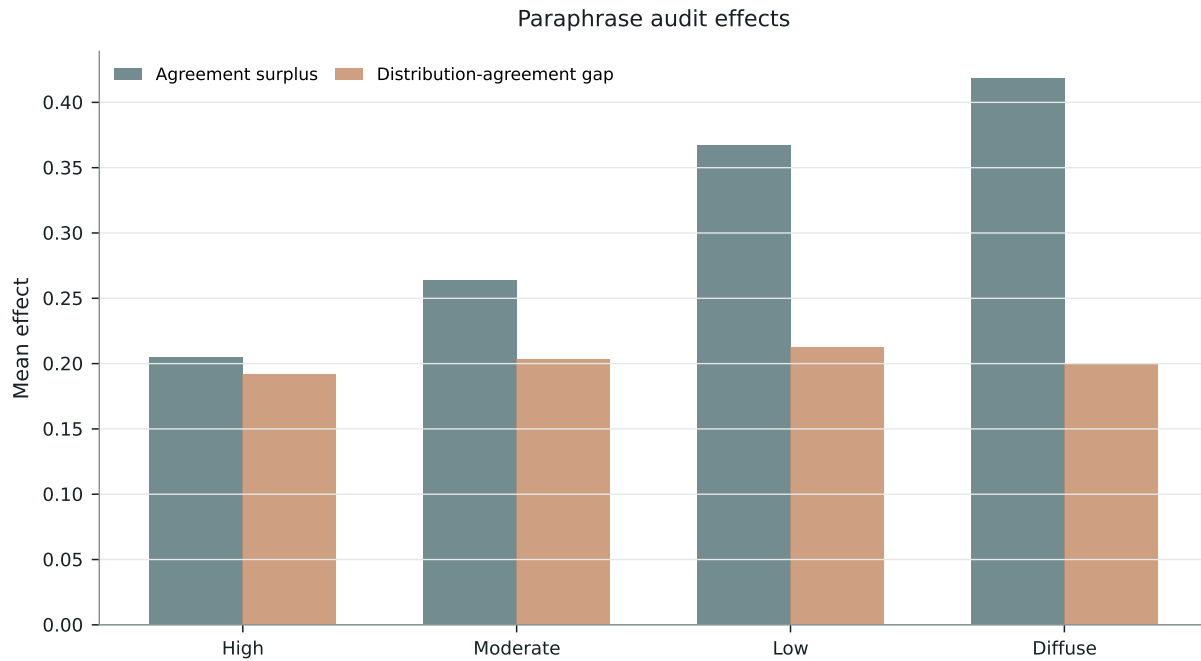


Figure 6: **Extended Data Fig. 2 | Aggregate paraphrase-audit effects.** Aggregate paraphrase-audit agreement surplus and distribution-agreement gap estimates preserve the direction of the main effects. The figure should be interpreted with the limited matched-coverage caveat stated in the Results and Supplementary Information.

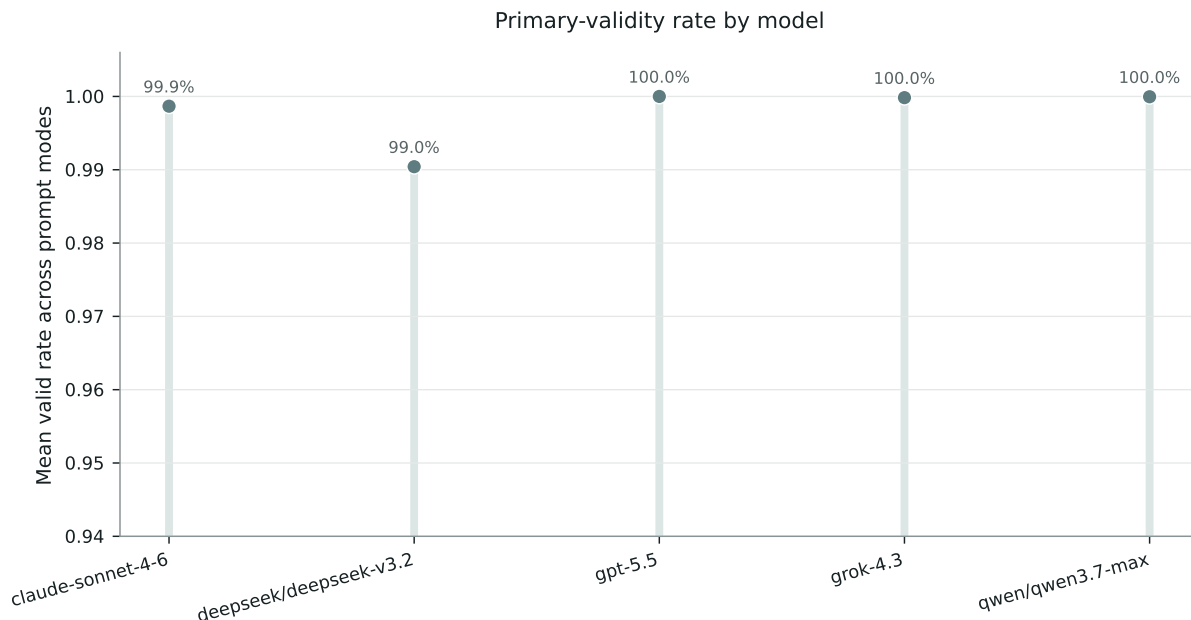


Figure 7: **Extended Data Fig. 3 | Validity by model and prompt mode.** Primary-valid output rates by model and prompt mode show high validity and limited invalid outputs in the frozen 50k target-scoped analysis.

Extended Data Table 1 | Validity and exclusion summary. Reports validity by model/mode, invalid status counts, minimum model/mode validity, refusal count, off-schema count, API errors, terminal failures and repair fields.

Supplementary Table 1 | Full model roster, API parameters, model-level endpoints and leave-one-model-out summaries. Provides model provenance, API-parameter reporting and cross-model consistency checks without making provider-family or model-family claims.

Supplementary Table 2 | Source-community distribution smoothing robustness. Reports raw-proportion, Jeffreys-smoothed and Laplace-smoothed source-community distribution sensitivity checks.

Supplementary Table 3 | Annotation and info robustness. Reports high-annotation-only, info-majority exclusion and high-info exclusion checks.

Supplementary Table 4 | Paraphrase audit. Reports paraphrase target calls, valid outputs, aggregate effects, confidence intervals, chosen-label stability and the matched-coverage caveat.

Supplementary Table 5 | Distribution quality and baselines. Reports distribution-quality diagnostics and uniform, global-base-rate and source-majority-oracle baselines.

Supplementary Table 6 | Normative certainty. Reports the secondary moral-certainty construct; moral certainty is not equivalent to estimated source-community agreement.

Supplementary Table 7 | Validity and exclusions. Reports validity and exclusion summaries and cross-references Extended Data Table 1.

Supplementary Table 8 | Model-ready secondary rows. Provides model-ready secondary rows as transparency files; these files do not contain reported mixed-effects model results.

Supplementary Notes. The Supplementary Information provides supporting methodological detail for prompt and schema reproducibility, source-community preprocessing, item allocation, model provenance, parsing and exclusions, endpoint definitions, bootstrap inference, robustness

checks, paraphrase limitations, distribution-quality diagnostics, normative-certainty secondary analysis and public/restricted materials boundaries.